

Andrew Dai

full-stack + AI dev with 10 consecutive hackathon wins and scaling Canada's largest high-school hackathon

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Education

University of Waterloo

Bachelor of Computer Science

Expected 2031

Waterloo, ON

Skills

- ◇ **Languages:** Python, JavaScript, Typescript, HTML, CSS, C++, C#, Java
- ◇ **Technologies:** Flask, React, Node.js, Next.js, Three.js, Tailwind, PyTorch, NumPy, Pandas, OpenCV, Redis, MediaPipe
- ◇ **Development Tools:** Git, MongoDB, GCP, AWS, LangChain, Claude Code, Docker, Node, Terraform, Postman

Experience

JAMHacks

Aug. 2025 – Present

Lead Organizer & Full-Stack Developer

Waterloo, ON

- ◇ Scaled Canada's largest high school hackathon up to **41.5% more attendees** in under a year on a **\$15,000 budget**
- ◇ Built and deployed jamhacks.ca and app.jamhacks.ca with **1000+ registered users** using **React + Node.js + MongoDB**
- ◇ Coordinating judging of **100+ projects** with engineers from **Google, Meta, Apple, & Microsoft**

Leading Aces Academy

Aug. 2025 – Dec. 2025

Contest Math Tutor

Markham, ON

- ◇ Prepared students for participation in national and international math contests such as **AMC, AIME, CSMC, & Euclid**

Open Source

Jun. 2024 – Present

Developer & Contributor

Remote

- ◇ Built an open source tool to sort similar photos by **perceptual hash**, designed to handle **large files (100GB+) in bulk**

Hackathons

Dec. 2022 – Present

Competitor, Judge, Mentor, Volunteer

North America

- ◇ Attended 18 hackathons with **10 consecutive wins**, shipping fully functioning products with users in **under 24 hours**
- ◇ Awarded **\$4000+ in prizes** building solutions for companies including **Databricks, MongoDB, Cohere, & Roboflow**

Projects

Semantic-Based Species Classifier – [Google Cloud Vision](#), [OpenAI Embeddings](#), [Flask](#), [React](#), [MongoDB](#)

- ◇ Built an AI animal classification pipeline that **supports 52.9% more species** than Google's SpeciesNet (3159 vs. 2066)
- ◇ Implemented a solution which matches a Google Vision description to **vector embeddings via semantic search** to generalize animals and **eliminate per-species training cost**
- ◇ **Designed to scale** with more vectorized species by using a general CV model instead of **training a specialized model**

Protosynthesis: Agentic Workflow Prototyping – [Gemini](#), [Moorcheh.ai RAG](#), [Flask](#), [MongoDB](#), [Supabase](#)

- ◇ Built a **tool-based agentic system** to build and test workflows using drag-and-drop tiles for API calls and logic
- ◇ Implements Moorcheh.ai **high-speed RAG** for **vector retrieval 6.6x faster** than Pinecone + Cohere solutions

Physiotherapy Game – [LangChain](#), [Terraform](#), [Google Cloud Platform](#), [Flask](#), [MongoDB](#), [OpenAI](#)

- ◇ Built a physiotherapy game which rewards proper exercise form, integrating motion tracking with a **Flask/MongoDB** backend, deployed on the **GCP Compute Engine** via **Terraform**
- ◇ Implemented an agent using **LangChain** and **OpenAI** to personalize feedback, awarded the **best use of Databricks**

Quantum Key Distribution – [Arduino](#), [Quantum Mechanics](#)

- ◇ Replicated the encryption **used by governments** for national security, storing data in the quantum superposition of light
- ◇ Demo **automatically transmits secure data** with motorized polarizers, impossible to intercept without corrupting data